Spreadsheets 2: IF, VLOOKUP, validation, templates, modelling, charts and macros

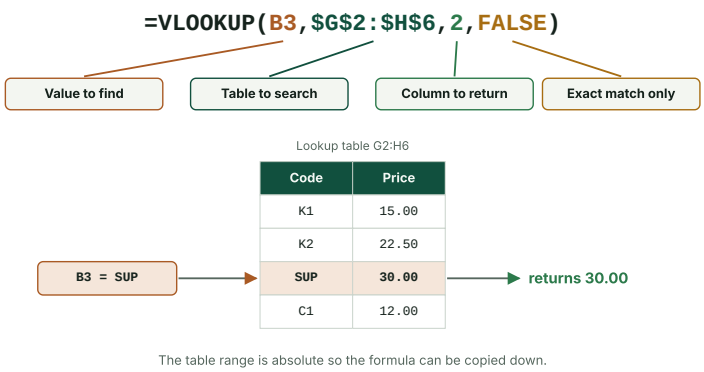
L12 FACT FILE

Specification: students should be able to

- Describe and use conditional formatting and data validation
- Describe and use templates, headers and footers, and importing data; create a simple template for others to use
- Use IF statements (including nested IF) and VLOOKUPs
- Use a spreadsheet for data modelling and what-if analysis
- Create, label and format charts
- Create simple macros

CCEA fact file: Spreadsheet 2

Key words: IF, nested IF, condition, VLOOKUP, lookup table, conditional formatting, data validation, range check, drop-down list, error alert, template, import, CSV, model, what-if, chart, legend, macro



The four parts of a VLOOKUP formula and the row it matches in a lookup table

Making spreadsheets work harder

Beyond simple sums, spreadsheets have features that make decisions, find information, check input and save repeated work. The IF function tests a condition and returns one result when it is true and another when it is false, so a sheet can show Pass or Fail, add a delivery charge, or choose between two exchange rates on its own. VLOOKUP searches the first column of a lookup table and returns matching information, such as a price, from the same row. Conditional formatting changes how a cell looks when a condition is met, so important values stand out. Data validation limits what can be typed into a cell, and an error alert asks the user to try again. Templates give a ready-made layout, and importing brings in data from other software without retyping. Spreadsheets can also be used as models to answer what-if questions, charts show results visually, and macros automate tasks that are done again and again. Each feature saves time, reduces errors or makes information easier to understand.

IF, nested IF and VLOOKUP

An IF function has three parts inside its brackets, separated by commas: the condition, the value if true and the value if false. =IF(C2>=50,"Pass","Fail") shows Pass for 50 or more. Text results need double quotation marks; numbers, cell references and calculations do not, so =IF(J2>=50,0,$K$1) gives free delivery on larger orders and otherwise uses the charge in K1. Read operators carefully: > means greater than, so a value equal to the boundary gives the false result. A nested IF places a second IF in the false part of the first, giving three possible results, such as Distinction, Merit or Fail. Test the highest boundary first and close every bracket. VLOOKUP has four parts: the value to look for, the table range, the column number to return from, and FALSE for an exact match. If the value is not in the first column of the table, the result is #N/A. The table range is usually absolute, such as $G$20:$I$24, so that the formula can be replicated down a column without the table moving.

Checking input and setting up for others

Conditional formatting applies a format, such as a red fill or bold font, only to cells that meet a rule. Because the rule is applied automatically, the formatting changes when the data changes, which is more reliable than colouring cells by hand. Data validation controls input at the point of entry. A range check might allow only whole numbers from 1 to 16; a list check gives a drop-down of allowed values, which is quicker and avoids spelling mistakes. An input message can explain what to enter, and an error alert pops up if invalid data is typed. Validation can only check that data is sensible, not that it is correct. A template is a pre-designed workbook with headings, formatting, formulae and validation already in place, so other people only add their data. Protecting the formula cells stops users breaking them. Importing uses a wizard to bring in data from a database table, text file or web page; the file may need to be in CSV format first.

Modelling, charts and macros

A spreadsheet model represents a real situation with input values and formulae that act as rules. Changing an input, such as a price, makes every dependent formula recalculate, so the user can see the effect straight away. This is what-if analysis. In a sports camp model, raising the price from £35 to £38 but losing 15 bookings changes profit from £540 to £600, which helps the organiser decide. Models let managers test many options quickly, cheaply and without risk, but the predictions are only as reliable as the data and rules used. Charts make results easier to understand at a glance and to show in presentations. Choose the type to suit the data: column or bar to compare categories, line for trends over time, pie for parts of a whole. A good chart has a title, axis titles with units and a legend when there is more than one data series. A macro records a sequence of actions so it can be replayed with one click, saving time and making sure repeated tasks are done the same way every time.

T Key terms from the mark schemes

IF function
A function that tests a condition and returns one value if it is true and another if it is false, e.g. =IF(A1>60,"PASS","FAIL").
Credited: U1 2022 Q3, 2023 Q4, 2025 Q6

Nested IF
An IF function placed inside another IF function so that more than two outcomes are possible, e.g. =IF(B2>=80,"Distinction",IF(B2>=65,"Merit","Fail")).
Credited: U1 2021 Q3, 2022 Q3

VLOOKUP
Vertical lookup: a function that searches for a value in the first column of a table and returns a value from a specified column in the same row. It updates automatically if the source data changes.
Credited: U1 2019 Q1, 2022 Q3, 2025 Q6

Lookup table
The range of cells searched by VLOOKUP; the values to be matched are in its first column.

Conditional formatting
Formatting (e.g. fill or font colour) that is applied automatically to cells only when a specified condition is met, e.g. hours worked over 40 highlighted.
Credited: U1 2019 Q1, 2025 Q6

Data validation
An automatic check when data is entered that it is sensible and within acceptable limits, e.g. a range check of 1 to 16. It cannot check the data is correct.

Error alert
A message displayed by data validation when invalid data is entered, asking the user to re-enter valid data.

Template
A pre-designed spreadsheet with layout, formatting and formulae set up, used as the starting point for new documents with the same format.
Credited: U1 2021 Q3, 2023 Q4

Importing data
Bringing data into a spreadsheet from another application or file (e.g. a CSV file exported from a database) without retyping it.

Data modelling / what-if
Using a spreadsheet to represent a real situation and changing input values to predict outcomes and see the effect of the change, e.g. changing a price to see the effect on profit.
Credited: U1 2019 Q8

Charts and graphs
Visual representations of data (e.g. column, line, pie) used to compare values, show trends or show parts of a whole; they can be copied into other software.

Macro
A small program that stores a sequence of actions so they can be carried out automatically with one click or shortcut, saving time on repeated tasks.
Credited: U1 2021 Q3

Exam focus

IF questions appear almost every year (2021 Q3, 2022 Q3(f), 2023 Q4(d), 2025 Q6(e)). Marks are given for the function name, a correct condition, both results in the right order and correct punctuation, so write the whole formula, including the equals sign, commas and quotation marks around text. Nested IF has been asked for three results (2021 Q3, 2022 Q3), and 2025 combined IF with an absolute reference. VLOOKUP appears as a multiple-choice question (2019 Q1(h)) and as an explain question (2022 Q3(g), 2025 Q6(f)). Examiners reported weak answers, so learn to explain each part: what is looked up, where it is searched for, which column is returned and why FALSE is used. Chart questions (2019 Q8(d)–(e), 2023 Q4(e)–(f), 2024 Q4) ask for the cell ranges used and the best chart type with a reason. Templates (2021 Q3, 2023 Q4(g)), validation and macros (2021 Q3) and what-if modelling (2019 Q8(f)) need a clear description plus a benefit linked to the scenario.

Unit 1 2025 Q6	[10]	Spreadsheet features, ranges, SUM, IF with absolute reference, lookup functions
Unit 1 2024 Q5	[10]	Spreadsheet functions, CPU purpose and components, immediate-access store
Unit 1 2024 Q4	[13]	SUM, subtraction, wrap text, absolute references, chart ranges and type
Unit 1 2023 Q4	[11]	Merge and centre, SUM, formulae, absolute references, IF, charts, template
Unit 1 2022 Q3	[12]	Currency, merge and centre, formulae, ranges, nested IF, VLOOKUP

? Check yourself

1. Cell D2 contains a test mark. Write an IF function that displays 'Merit' if the mark is 70 or more, otherwise 'Pass'. [2]
2. Explain what the function =IF(A1>60,"PASS","FAIL") does. State the result if A1 contains 60. [3]
3. Cell B2 contains a student's mark. Write a nested IF function for C2 that displays "Distinction" for a mark of 80 or more, "Merit" for a mark of 65 or more and "Fail" otherwise. [3]

Answers: 1. =IF(D2>=70,"Merit","Pass") – correct condition D2>=70 (1) / both results in the right order with quotation marks (1) 2. Checks whether the value in A1 is greater than 60 (1) / if true displays PASS, otherwise displays FAIL (1) / A1 = 60 gives FAIL because 60 is not greater than 60 (1) 3. =IF(B2>=80,"Distinction",IF(B2>=65,"Merit","Fail")) – first condition and Distinction (1) / nested IF with B2>=65 and Merit (1) / Fail as final false value with correct brackets and quotes (1)